Supply Chain Management and Logistics

Group Project Guidelines

Overview

You are working as a **supply chain analytics team**.
In this project, your group is required to:

- Identify a real-world supply chain problem
- Find and collect relevant datasets
- Use SQL to integrate and analyse the data
- Provide data-driven insights and recommendations

Unlike the individual assignment, the dataset is **not provided**.
Each group must independently identify and prepare their own data.

Business Context

As a data analyst, your team's role is to support decision-making in the supply chain and operations of the platform. The company is interested in improving performance in areas such as:

- ✧ operations optimisation
- ✧ delivery and fulfilment
- ✧ inventory and product strategy
- ✧ supplier or seller management
- ✧ customer behaviour
- ✧ logistics and geographic efficiency

Your analysis should aim to support **practical business decisions**, not just describe data.

Project Requirement

Each group must choose **ONE core supply chain problem** and analyse it in depth.
Examples of core problems include:

- How can the company improve inventory health and replenishment decisions?
- Where are the main inefficiencies in delivery and fulfilment?
- Which sellers create the greatest operational risk?
- Which product categories require different supply chain strategies?
- How can customer or regional patterns support logistics optimisation?
- Any other supply chain related problem approved by the instructor

A strong project should not be a collection of unrelated SQL queries.
It should show a **clear analytical logic**:
business overview → problem diagnosis → deeper analysis → recommendation

Supply Chain Management and Logistics

Required Project Structure

1. Problem Definition

Clearly define the business problem your group is investigating.

You should explain:

- What problem are you solving?
- Why does this problem matter?
- What business decisions should this analysis support?

---

2. Data Selection and Data Preparation

Explain the data used in your project.

You should include:

- CSV file or files that were used
- How the tables were linked if there were multiple csv files
- What data was removed or kept
- Any cleaning or transformation performed
- Any limitations in the data

---

3. Business Overview Analysis

Before moving to detailed analysis, each group must first develop an overall understanding of the business.

This section should include summary analysis such as:

- overall order volume
- sales or revenue trend over time
- number of customers, products, sellers, or categories
- general business scale and pattern

This section should help the reader understand the broader business context before focusing on the chosen problem.

---

4. Diagnostic Analysis

This section should investigate the chosen problem from multiple perspectives.

Groups should conduct at least 2–3 focused analyses to diagnose the issue.

For example:

- by category
- by seller
- by region
- by warehouse
- by customer segment
- by time period

Each analysis should answer a clear business question, rather than simply showing a table of numbers.

5. Deep-Dive Analytical Section

Each group must include at least one deeper analytical section that goes beyond basic descriptive analysis.

This may include, for example:

- ABC analysis
- inventory turnover analysis
- supplier scoring model
- customer segmentation
- delay risk classification
- geographic efficiency comparison
- safety stock / reorder point logic
- ranking, scoring, or classification models
- other advanced SQL-based analytical logic

This section should demonstrate that the group can use SQL not only to retrieve data, but also to support business evaluation and decision-making.

---

6. Results and Visualisation

Groups must present the key results clearly.

Requirements:

- include at least 2 meaningful visulization
- charts should support the argument of the report
- explain what each chart shows
- explain why the result matters

Visualisations should not be only decorative. They should help communicate important findings.

---

7. Key Insights

Summarise the main findings of your project.

You should explain:

- What patterns did you discover?
- What problems or opportunities were identified?
- Which findings are most important?
- Why do these findings matter to the business?

This section should focus on interpretation, not just repeating the results.

---

8. Recommendations

Provide at least 3 actionable recommendations based on your analysis.

For each recommendation, explain:

- What should be done
- What evidence supports this recommendation
- Who should act on it (if relevant)

- What business benefit is expected
- Any possible risk or limitation

Strong recommendations should be specific, realistic, and clearly linked to the analysis.

9. Individual Contribution Statement

At the end of the report, each group member must briefly state:

- what part(s) they worked on
- what type of contribution they made
- how they supported the final project

This section is required for fairness in group assessment.

Deliverables

Students must submit the following:

1. Report Length Guidelines

The main report should be approximately:

- 10–20 pages (excluding cover page, references and appendix)

As a reference, the report is expected to be around:

- 2500–4000 words, but this is not a strict limit

The focus should be on:

- clarity of analysis
- logical structure
- depth of insights

rather than length.

Appendices may include:

- SQL scripts
- additional tables
- extended outputs

There is no strict page limit for appendices, but they should be relevant and clearly organised.

Page size: A4
Margin: 2.54 cm per Side
Cover Page (1 page)

- Project Title
- Details of the Group
  - ◊ Students' IDs
  - ◊ Names
  - ◊ Course name
  - ◊ Section Number

Font: Times New Roman, 12 pt
Spacing: Double Spacing

Submit E-copy only in PDF format
Submit hard copy to TA or the instructor
File name format: GroupX_SectionY.pdf (e.g., *Group1_1001.pdf*)

2. Supporting Evidence (Required)
Must include:

- SQL scripts
- final combined table
- dataset source description

3. Presentation Slides (PowerPoint/Google Slides)
Slides should provide a concise summary of your work:
Submit your soft copy with your report and a hard copy when you present your work

- **Presentation time: 15 minutes per group and 5~10 mins Q&A.**
- **Order of groups: decided by a lucky draw.**

Assessment Criteria

Criteria	Weight
Problem framing and business relevance	10%
Data preparation and relational design	10%
SQL correctness and technical depth	25%
Diagnostic analysis quality	20%
Deep-dive analysis / model application	15%
Insights and recommendations	15%
Clarity, visualisation, and presentation	5%

Supply Chain Management and Logistics

FAQ

1. Do we need to include SQL queries in the report?

No. You do not need to include all SQL queries in the main report.
However, you must submit your SQL scripts as supporting evidence.
Your report should clearly explain:

- how the data was combined
- what analysis was performed
- how the results were obtained

2. Can we use only one dataset?

Yes. Some datasets might come as one only but you need to make sure the datasets offer enough layers for depth analysis.

3. What is considered a "good dataset"?

A good dataset:

- has clear business meaning
- contains multiple variables (e.g. time, category, metrics)
- allows relational analysis (can be joined)
- supports meaningful business questions

Avoid:

- very small datasets
- purely descriptive or static data

4. What is considered a good analysis?

A good analysis:
combines data from multiple tables
explains patterns and relationships
goes beyond simple descriptions

5. Can we use Excel or Power BI instead of SQL?

No.
SQL must be used as the core analysis tool.
Other tools (Excel, Power BI, Tableau) may be used for visualisation only.

6. Can I use AI tools (e.g., ChatGPT) to help write the report?

AI tools may be used to help with **grammar, wording, or editing**. However:

- The reflections must be **based on your own learning experience**.
- Reports that appear entirely AI-generated and not connected to the exercises may receive reduced marks.

Supply Chain Management and Logistics

AI Usage Policy

All submissions will be checked using Turnitin, including its AI writing detection tool.

As a general guideline:

- AI detection scores **below 20% are generally considered low and may occur naturally in academic writing**.
- Reports with **AI indicators above 30% may be subject to further review**.
- If necessary, the instructor may ask students to **clarify or explain parts of their report**.

Students should ensure the report reflects **their own understanding and learning experience**.